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What to look for?

Your test system should be based on open measurement platforms which have the following characteristics:

- Wide I/O connectivity options for integrating multiple test techniques like Flying Probe ICT, Boundary Scan, Optical Inspection & Functional Test into one system.
- Modularity to meet the ever-changing test requirements
- Open standards for third-party connectivity
- High channel count for probing high-component-density circuit card assemblies (CCAs)
- Efficient test management software for increased throughput and test reuse

ful for analogy testing as well. Since these are integrated in the same digital platform, you are able to effectively test digital, mixed logic, analogue or mixed-signal assemblies.

Integrated intelligence. Integrated intelligence is taking ICT to a whole new level. There are learn and compare features included where the system learns from the golden board and then compares the results. Also, some can learn from past test results and perform a few additional tests, if needed, to produce useful and comprehensive reports.

Increased precision with rotational movement of flying probes

The robotic testers have added a lot of automation. These testers, or the flying probes or the automated probes in ICT systems, can go as close as 20 mils/500 microns between two pins. Complex PCBs sometimes have components that are soldered at certain angles to the PCB plane. So, for high accuracy, the probe should also be able to move at an angle.

Previously probes could move only in x-y direction, but now systems include range expansion rotational movement where you can rotate the probe at dif-

ferent angles, thereby allowing you to go very close to the adjacent pins and increasing the testing accuracy.

QTouch 1248 Flying Prober from QmaxTest is based on the technology of hands-free probing which can probe both sides of the board due to its vertical design. It provides scalability from one to eight probes with a maximum of

four probes on each side of the board. Apart from X, Y and Z axes, probes can move in theta direction up to an angle of 36 degrees.

Acculogic's Flying Scorpion FLS-980Dxi offers programmable probe angles of +6° to -6° with ± 0.1 micron resolution for testing.

Software Help

With the latest software tools, application engineer's job has been totally reduced. With graphical user interface (GUI), you perform all the tests in a graphical way. There is no need to write codes for complex circuits. The GUI provides flexibility to the system and assists test environments for different production cycles.

You can also define your own test program by creating different steps that can be sequenced in an automated sequence. You can give your own conditions and put conditional jumping between different steps. For example, if a particular test is passed then jump to a specific step, otherwise to the other one.

Also, you can protect your test conditions with passwords and ensure another user cannot change them; he can only run the test. You can maintain a record of how many boards were tested

and how many failed, and have all the relevant reports of yield rate or efficiency.

For quick debugging and diagnostics, where you want to quickly test any particular area of the PCB, software allows you to perform these localised tests without writing any test sequence. So, if you want to test any particular IC or a few ICs or any other component, you can just make interface for the device under test using interactive GUIs and perform the required tests.

Drawback. The only drawback of the ICT software tools is that they are unique and proprietary for each system. Ganapathy of QmaxTest says, "Every tester has its own driver and capabilities, and so every tester should be directly sequenced with a set of instructions. If someone is thinking of developing a software, they can see the software function and develop one but, however, when you buy any ICT system, the proprietary software only will work."

Life of an ICT system

Testing at component level will remain for years, but with technological advancements these components are getting smaller and making the circuits more and more complex. These advancements have caused older techniques to fail, and will continue to do so with the coming systems as well. Is there a solution?

Ganapathy explains, "An important consideration while selecting any solution should be the number of test points and the associated scalability. A manufacturer should first analyse how many board types are to be tested, and how many maximum test points are required, and then, finally, what scalability is available with the solution."

He adds, "Today you might not require a certain number of test points, but at least the equipment should have the capability to add more in future when there is a requirement. For these reasons there are more of open architecture systems coming up, so that as the requirement increases you are able to increase the channel capacity or add new test methodologies as well." ●

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